

Dynamotype Atlas

Overview

This atlas provides a comprehensive classification of 16 seizure dynamotypes based on the Saggio et al. model. It includes descriptions of the different classes of seizures, 5 sample state-space trajectories, 3 noise levels, and their onset and offset bifurcation dynamics. The atlas also introduces a novel hybrid method for trajectory generation that refines previous classifications.

Note on `seizure_atlas.mat`:

You cannot download the `seizure_atlas.mat` file through the repository's "Download ZIP" option, as it will not be included. You must either clone the repository directly using Git or download the file individually from the repository.

Structure

The document contains:

- A detailed explanation of three trajectory generation methods:
 - **Hysteresis Bursters**
 - **Slow-Wave Bursters**
 - **Piecewise Slow-Wave Bursters**
- Individual descriptions of each seizure class, including bifurcation types and state-space representations.
- Illustrations of 5 state-space trajectories and corresponding time-series behavior.

Supplementary Code

This atlas includes code to generate figures and simulate seizure trajectories. The code allows users to:

- Replicate the 330 simulated seizures found in the atlas.
- Simulate seizures based on different onset and offset bifurcations.
- Generate visualizations of state-space paths and time-series representations.

Usage

1. **Data Preparation:** Ensure that any required datasets or parameter files are available.

2. **Running the Simulations:** Execute the provided scripts to generate seizure simulations.
3. **Visualizing Results:** Use the plotting functions to compare different seizure classes.

Supplementary Seizures

This atlas includes a mat file of all 330 seizures named seizure_atlas.mat. Below is a table that shows the specified dynamotype within indices 1-330.

Index Range	Dynamotype
1–15	SN/SNIC
16–30	SN/SH
31–45	SN/SH variant 1
46–60	SN/SH no DC
61–75	SN/Sup
76–90	SN/Sup variant 1
91–105	SN/Sup variant 2
106–120	SN/FLC
121–135	SNIC/SNIC
136–150	SNIC/SH
151–165	SNIC/Sup
166–180	SNIC/Sup variant 1
181–195	SNIC/FLC
196–210	Sup/SNIC
211–225	Sup/SNIC variant 1
226–240	Sup/SH
241–255	Sup/Sup
256–270	Sup/FLC
271–285	Sub/SNIC
286–300	Sub/SH
301–315	Sub/Sup
316–330	Sub/FLC